

Sumitomo Drive Technologies
Always on the Move

Cyclo[®] 6000

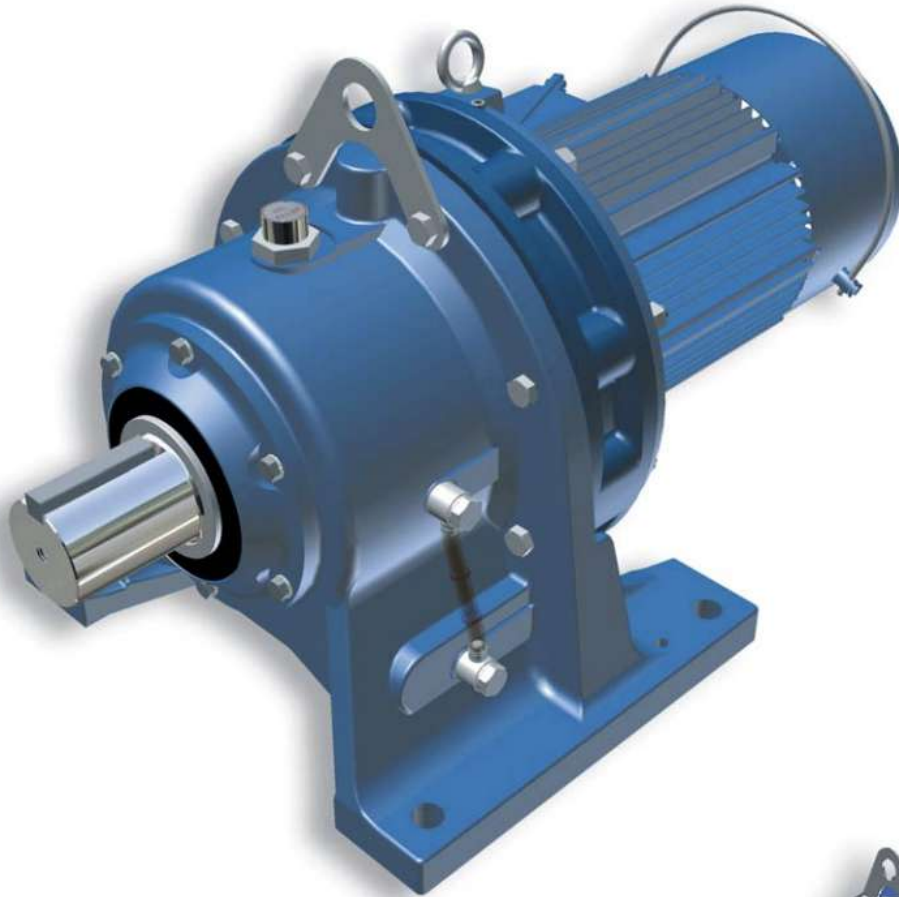
Gearmotors



CATALOG 04.601.50.004

Cyclo® 6000

Superior design, powerful performance



- ▶ The Cyclo® 6000 is also available as a Speed Reducer

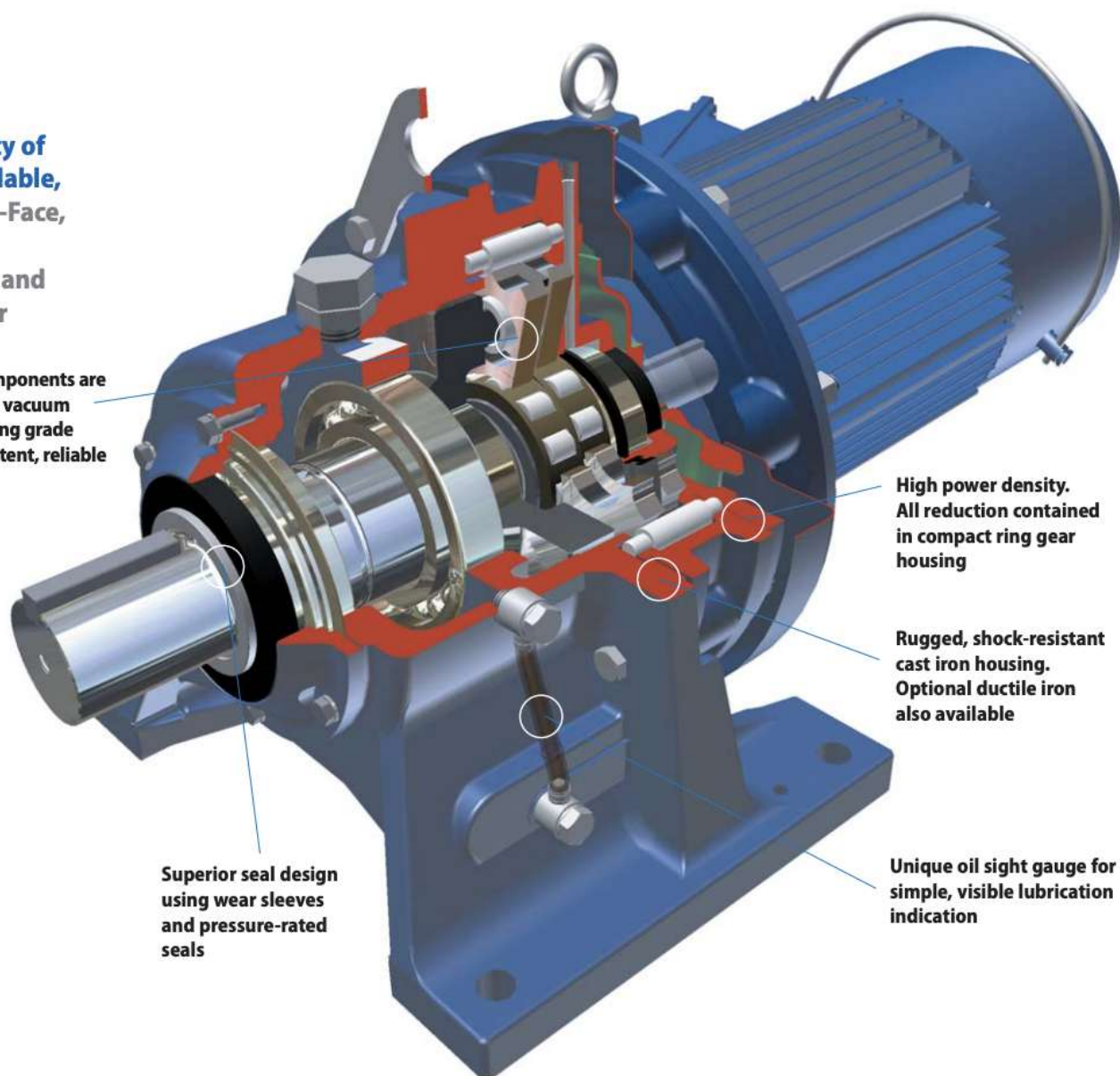
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www.sumitomodrive.com

Cyclo® 6000

- **Wide variety of inputs available, including C-Face, Free-Shaft, Gearmotor and Brakemotor**

All rotating components are fully hardened, vacuum degassed bearing grade steel, for consistent, reliable performance



Superior seal design using wear sleeves and pressure-rated seals

High power density. All reduction contained in compact ring gear housing

Rugged, shock-resistant cast iron housing. Optional ductile iron also available

Unique oil sight gauge for simple, visible lubrication indication



Unmatched Reliability, Exceptional Performance

- Cyclo® speed reducers and gearmotors are designed to withstand shock loads exceeding 500% of their ratings



Highly Reliable, Torque Dense Cycloidal Gearmotors

Product Description

The Sumitomo Cyclo® drive is **unsurpassed by any other inline drive** available in the market today. **Cyclo®'s unique cycloidal design** has advantages superior to speed reducers using common involute gears. Cyclo® components operate in compression, not in shear. Unlike gear teeth with limited contact points, a Cyclo® has two thirds of its reduction components in contact at all times. Cyclo® speed reducers and gearmotors are **designed to withstand shock loads exceeding 500%** of their ratings, and provide exceptional performance, reliability and long life in the most severe applications.

Features & Benefits

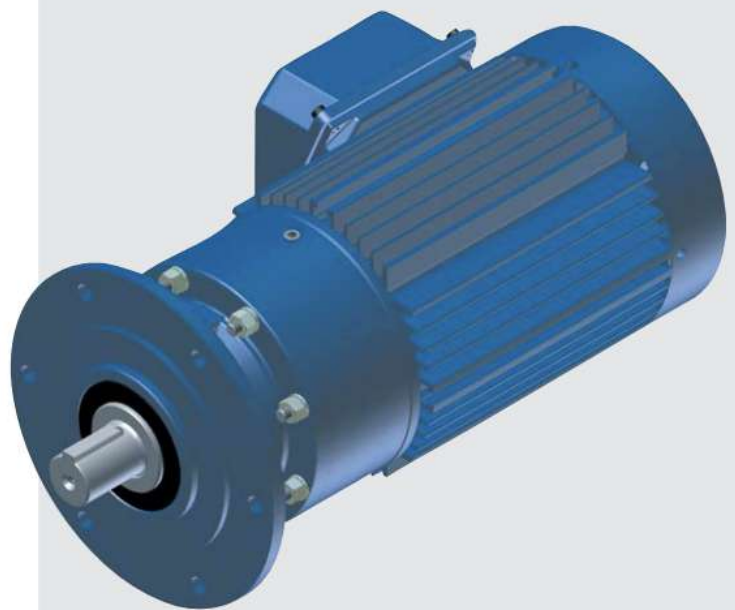
- **Highest overload capacity**, exceeding 500%
- **Exceptional life** with a 24 month warranty
- **High efficiency**, even at high reduction ratios
- **Versatile**, available as gearmotor or inline speed reducer
- Ideal for **severe, high shock** applications
- Optional grease lubrication for low maintenance

Specifications

Sizes:	23 models (5 lbs to 5000 lbs)
Torque Rating:	55 to 603,000 lb in
HP Rating:	.10 to 235 HP
Ratio Range:	3:1 to 119:1 (single) 121:1 to 7569:1 (double) 8041:1 to 658,503:1 (triple)
Mounting:	Foot, Flange, Face Mount

- ▶ **Sumitomo's Cyclo® 6000 is extremely torque dense and is available as an inline gearmotor or speed reducer**

- Connection free design
- Rugged forged output shaft
- Direct acting brake option
- Unmatched durability
- Foot and Flange Styles



Applications

- Conveyors
- Food Machinery
- Mixers
- Automotive Plants
- Recycling Machines
- Poultry Plants
- Sawmills and Wood Mills
- Wastewater Treatment
- Steel Mills
- Construction Equipment
- Paper Mills
- Processing Plants

FAQs

How do I select a Cyclo® gearmotor?

Selection is based on the motor horsepower and/or torque requirements at the output shaft. The Cyclo® gearmotor has particularly high efficiencies over a wide range of reduction ratios, which frequently permits the use of reduced input power requirements (smaller HP motor) without sacrificing output shaft torque. The selection procedures in this catalog will guide you in choosing the most efficient gearmotor for your application.

What information do I need to get started in the selection process?

To select the proper gearmotor for your application, you will need to know:

- Application: type of driven machine
- Hours of operation per day
- Motor horsepower (HP) and speed (RPM)
- Mounting position

If there are any special environmental factors or operation requirements, they must also be noted. This information will be important in determining the Service Factor of your application.

What are Service Classes and how are they used?

In general, gearmotors are rated for the specific conditions and operating requirements of the application by the use of AGMA-defined Service Classes. There are three AGMA Service Classifications for gearmotors: uniform (I), moderate shock (II) and heavy shock (III) (pages 2.4-2.5) The Service Classes are used in the product selection process to adjust for the specific conditions and operating requirements of your application.

What do I do if my application has particularly severe operating conditions?

The standard ratings for Cyclo® are based on 10-hour daily service under conditions of uniform loads (equivalent to AGMA service Class I). By following the product selection process, you will determine and apply the Service Factors to compensate for longer periods of operation and/or severe operating conditions.

How can I be sure that the gearmotor can withstand periodic excessive overloads?

Cyclo® Gearmotors provide 500% momentary intermittent shock load capacity. Planetary gearmotors can accommodate 300% momentary shock loads. For applications with shock loads greater than 500%, consult an SMA Application Engineer.

What are the standard input speeds?

In general terms, the speeds are 1750 and 1165 RPM. When non-standard input speeds are used, the horsepower and torque ratings also vary.

What thermal capacity limitations does the Cyclo® have?

The Cyclo® gearmotor, by virtue of its smooth, almost frictionless operation (unlike traditional helical gears), has a thermal rating that far exceeds its mechanical capacity and all but eliminates the conventional limitations due to heat under normal ambient conditions.

What inverter turn-down ratios are available?

A 2:1 ratio is standard with Class F insulation. A 10:1 ratio is available for large frame inverter duty motors. A 1000:1 ratio is available in a C-face configuration.

Common Configurations



Single Reduction,
Horizontal Foot Mount
Brakemotor



Single Reduction, Flange Mount



Single Reduction, V-Flange Mount



Double Reduction

Standard Specifications

3-Phase Integral Motor	Standard Specifications		Standard Specifications with Built-In Brake	
	Capacity Range:	1/8 HP ~ 75 HP, 4P 20 ~ 75 HP, 6P	1/8 HP ~ 15 HP, 4P: FB Brake 20 HP, 4P: CMB Brake 25 HP ~ 40 HP, 4P: ESB Brake	
	Enclosure:	Totally enclosed fan cooled type (1/8 HP, 4P Totally enclosed non ventilated)	Totally enclosed fan cooled type (1/8 HP, 4P Totally enclosed non ventilated)	
	Power Supply:	230/460 Volts, 60 Hz 575 Volts, 60 Hz	230/460 Volts, 60 Hz 575 Volts, 60 Hz	
	Insulation:	3/4 ~ 30 HP: Class B 40 HP: Class F	3/4 ~ 20 HP: Class B	
	Time Rating	Continuous	Continuous	
	Standards	UL, CE, CSA are available	Please consult factory	
	Note:	<i>Dimensions for 575 Volt and CSA units may be different than those specified in Section 4, please consult factory.</i>		
Reducer	Reduction:	Involute crowned tooth profile for ratios 3 and 5:1. Internal planetary gear mechanism with trochoidal curved tooth profile for ratios 6:1 and higher.		
	Lubrication:	Grease or oil lubricated models available.		
	Seals:	Nitrile material, dual lipped, double output seals available.		
	Material:	Rugged cast iron or ductile housings.		
	Paint Color:	Blue, Muenters color number 6.5PB 3.6/8.2		
	Note:	<i>Ratings listed in Section 3 may be altered if models normally designed for oil lubrication are special ordered for grease lubrication; please consult factory.</i>		
Ambient Conditions	Installation Location:	Indoors (Minimal dust and humidity)		
	Ambient Temperature:	14°~104° F (-10° ~ 40° C)		
	Ambient Humidity:	Under 85%		
	Elevation:	Under 3,281 ft. (1000 meters)		
	Atmosphere:	Well ventilated location, free of corrosive gases, explosive gases, vapors and dust.		

Shaft Rotation

On single reduction Cyclo® gearmotors, ratios 3 through 119, the slow speed shaft rotates in a reverse direction to that of the motor shaft.

On double reduction units, ratios 104 through 7569, both the motor and the slow speed shaft rotate in the same direction.

Input Speeds

In general terms, the standard input speeds of single reduction units are 1750 and 1165 RPM. When non-standard input speeds are used, the horsepower and torque ratings will also vary.

Thermal Capacity

The Cyclo® gearmotor's smooth, almost frictionless operation all but eliminates the conventional limitations due to heat. In all sizes, Cyclo® gearmotors have thermal ratings that exceed their mechanical capacity.